Clinical Study Summary Report (CSSR)
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1. Administrative & Study Identification

Full Study Title: A Phase 1, Multicenter, Open-Label Study Of CC-97540 (BMS-986353), CD19-Targeted Nex-T Chimeric Antigen Receptor (CAR) T Cells, in Participants With Severe, Refractory Autoimmune Diseases: Systemic Lupus Erythematosus, Idiopathic Inflammatory Myopathy, Systemic Sclerosis, or Rheumatoid Arthritis (Breakfree-1)
Short Title/Acronym: Breakfree-1
Protocol Number: CA061-1001
NCT Number: NCT05869955
Posting ID: 686384222855272186 **Sponsor/Company Name:** Bristol Myers Squibb (Juno Therapeutics, Inc., a Bristol Myers Squibb Company) **Investigational Product:** CC-97540 (BMS-986353), an investigational, autologous CD19-targeted NEX-T™ CAR T cell therapy
Phase of Development: Phase 1
Trial Status: Recruiting **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To establish the safety, tolerability, and preliminary efficacy of the leukapheresis, lymphodepletion, and infusion of the study intervention (CC-97540) in participants with severe, refractory autoimmune diseases. **Secondary Objectives:** To evaluate the pharmacokinetics of CC-97540, the depth and duration of B cell depletion, efficacy measures of disease remission/activity, and to determine the recommended Phase 2 dose (RP2D).

2.2 Background & Rationale Despite extensive efforts to develop novel therapies for Systemic Lupus Erythematosus (SLE), Systemic Sclerosis (SSc), Idiopathic Inflammatory Myopathy (IIM), and Rheumatoid Arthritis (RA), an unmet need remains for treatment-refractory patients suffering from recurrent disease and long-term treatment-related toxicities. Failure to respond to systemic glucocorticoids and immunosuppressive drugs indicates a poor overall prognosis. This study aims to evaluate CC-97540's potential to drive an "immune reset," leading to sustained, drug-free disease control in these highly refractory populations.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose escalation/expansion) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target Enrollment: 270 participants **Number of Sites:** Multicenter (International / US) **Estimated Study Start:** 13-Sep-2023 **Estimated Primary Completion:** 29-Aug-2028

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Diagnosis of the respective severe autoimmune disease fulfilling established classification criteria (e.g., 2019 EULAR/ACR for SLE, 2013 EULAR/ACR for SSc). 2. Active disease at screening (e.g., for SLE, recent $\ge$ 1 major organ system with a BILAG A score). 3. Inadequate response to standard-of-care treatments (e.g., for SLE, inadequate response to glucocorticoids and at least 2 immunosuppressive therapies used for $\ge$ 3 months each).

4.2 Exclusion Criteria 1. Diagnosis of drug-induced SLE rather than idiopathic SLE. 2. SLE overlap syndromes including, but not limited to, rheumatoid arthritis, scleroderma, and mixed connective tissue disease. 3. Present or recent clinically significant central nervous system (CNS) pathology within the last 12 months.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: A single infusion of BMS-986353 (CC-97540) administered at either 10×10^6 (DL1) or 25×10^6 (DL2) CAR+ T cells. **Route of Administration:** Intravenous (IV) infusion **Duration of Treatment:** Single infusion, followed by extensive safety and efficacy monitoring for up to two years.

5.2 Concomitant & Prohibited Therapy Lymphodepleting Regimen: Includes Fludarabine and cyclophosphamide (ATC Code: L01AA01; Trade Names: Cytoxan, Endoxana). **Permitted:** Standard supportive care post-infusion; Tocilizumab or other approved agents for the management of Cytokine Release Syndrome (CRS) if required. **Prohibited:** Disease-directed therapies and chronic immunosuppressants must be discontinued prior to the lymphodepletion conditioning regimen.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Eligibility Assessment, Baseline Disease Activity Scoring
Baseline	Day -7 to 0	Leukapheresis, Lymphodepletion (Fludarabine/cyclophosphamide), Washout of prior therapies
Intervention	Day 0	Single IV infusion of CC-97540 (BMS-986353)
Interim	Weeks 1 to 24	Pharmacokinetics, Safety Monitoring (CRS, ICANS), B-cell depletion tracking, Efficacy Assessment (Week 24)
End of Study	Up to Year 2	Long-term safety follow-up, evaluation of sustained drug-free disease control

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. Number of participants with treatment-emergent adverse events (AEs) in each indication. 2. Number of participants with serious AEs (SAEs) in each indication. 3. Number of participants with AEs of special interest (AESI) in each indication.

7.2 Secondary Endpoints 1. Proportion of participants achieving definition of remission in SLE (DORIS) remission. 2. Proportion of participants achieving Lupus Low Disease Activity State (LLDAS). 3. Change in proteinuria measured by urine protein creatinine ratio (UPCR). 4. Pharmacokinetics of CC-97540 and depth/duration of B-cell depletion.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Intensive tracking of cell therapy-specific toxicities, including Cytokine Release Syndrome (CRS), Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS), cytopenias, and inflammatory AEs. **Serious Adverse Events (SAEs):** Evaluated strictly to establish the safety profile and the Recommended Phase 2 Dose (RP2D). **Data Monitoring Committee (DMC):** Internal/External safety review committees utilized to analyze dose escalation data (DL1 vs. DL2).

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients receiving $\ge$ 1 dose of study treatment) and Efficacy Evaluable Set. **Sample Size Justification:** Bayesian optimal interval design used to guide dose escalation and the selection of the Recommended Phase 2 Dose (RP2D). Estimated overall target $N=270$. **Handling of Missing Data:** Standard approaches for early-phase cellular therapy trials (e.g., censoring at last available follow-up for time-to-event endpoints).

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite and remote monitoring tailored for gene-modified autologous cellular therapies and their unique safety requirements. **Audit Trail:** eCRF locking, tracking of leukapheresis material chain of custody, and clinical event adjudication forms.

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11. Study Identifiers & Governance

Primary ID: CA061-1001 **Registry Number:** NCT05869955 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** Breakfree-1 **Official Regulatory Title:** A Phase 1, Multicenter, Open-Label Study Of CC-97540 (BMS-986353), CD19-Targeted Nex-T Chimeric Antigen Receptor (CAR) T Cells, in Participants With Severe, Refractory Autoimmune Diseases: Systemic Lupus Erythematosus, Idiopathic Inflammatory Myopathy, Systemic Sclerosis, or Rheumatoid Arthritis (Breakfree-1)

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12. Clinical Framework (Autoimmune Specific)

Disease Areas: Systemic Lupus Erythematosus, Idiopathic Inflammatory Myopathy, Systemic Sclerosis, Rheumatoid Arthritis (Arthritis) **MeSH Code:** D001168 (Arthritis) **Preferred UMLS Name:** CAR T-Cell Therapy for Autoimmune Diseases **Diagnosis Threshold:** Active disease despite standard of care (e.g., recent $\ge$ 1 major organ system with a BILAG A score for SLE) **Medical Methodology:** Autologous CD19-directed CAR T cell therapy (immune reset) **Optimization Target:** Complete peripheral B-cell depletion and clinical symptom resolution

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13. Study Procedures & Methodology

Management Pathway: Leukapheresis, Lymphodepletion (Fludarabine/cyclophosphamide), and CAR-T cell infusion **Education Protocol (HCP):** Training on identification and management of cell therapy toxicities (CRS, ICANS) **Education Protocol (Patient):** Consenting and education on the risks of CAR-T therapy, lymphodepletion, and the necessity of long-term follow-up **Quality Audit Mechanism:** Tracking of CAR T cell expansion, pharmacokinetic monitoring, and evaluation of sustained drug-free disease control

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14. Observation & Follow-Up Schedule

Total Duration: Up to 2 years for primary follow-up **Onsite Visit Cadence:** Frequent early visits (Days 1-28) followed by Weeks 12, 24, and routinely every 3-6 months **Remote Monitoring Frequency:** Routine safety calls between onsite visits **Checklist Review Interval:** Regular assessment of disease activity scores (e.g., SLEDAI-2K, PGA) at key milestones

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	**[DD-MMM-YYYY]**
Clinical Operations Lead	**[Name]**	___	***[DD-MMM-YYYY]***
Lead Statistician	***[Name]***	_____	[DD-MMM-YYYY]